

## Week 10 — Preventive Maintenance, Case Studies & Certification

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**Sessions:** 1 (Theory) · 2 (Assessment Lab) · **Total:** 4 hours

**Primary LOs:** LO-09, LO-12

**Hands-on target:** Session 2 ≈ 70% practical/assessment

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### Instructor Preparation Guide (Complete Day Before)

#### Pre-session setup

- [ ] Finalize case-study cart assignments (from Weeks 7–9 work where possible)
- [ ] Prepare final practical carts with unique faults (one per student or per pair if class large)
- [ ] Print final practical rubric (`final_assessment/practical_skills_rubric.md`) — one per student
- [ ] Print written final exam + keep answer key secure (`final_assessment/written_final_exam.md`)
- [ ] Print certificates (`final_assessment/certificate_template.md`) — fill names after scoring
- [ ] Prepare feedback forms
- [ ] Stage PM checklist handouts for Session 1 exercise
- [ ] Schedule time: presentations → practical → written (or written first if quieter)

#### Final practical fault bank (examples)

| Code | Fault                                        | Max time |
|------|----------------------------------------------|----------|
| FP1  | High-resistance battery cable / corroded lug | 45 min   |
| FP2  | Solenoid contact failure                     | 45 min   |
| FP3  | Throttle signal fault                        | 45 min   |
| FP4  | Charger interlock open                       | 45 min   |
| FP5  | Controller flash code + related cause        | 45 min   |

#### Grading prep

- Calculator for weighted scores per Program Guide
  - Attendance sheet finalization
  - Critical safety fail = no certificate until makeup demonstrated
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## Week 10 Learning Objectives

1. Design and implement effective PM programs for golf cart fleets.
2. Present a complete diagnostic case study with findings and recommendations.
3. Demonstrate overall competency via final practical and written assessments.

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## Session 1 — Theory & Demonstration (2 hours)

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**Materials needed:** handouts/pm\_checklists.md, labs/W10\_pm\_plan\_worksheet.md; career pathway one-pager; program review question cards; case study rubric copies.

### Timed Agenda

| Time | Min | Activity                                                             | Instructor Notes                      |
|------|-----|----------------------------------------------------------------------|---------------------------------------|
| 0:00 | 5   | Safety + Week 10 roadmap (PM → presentations → finals)               | Reduce anxiety; clarify pass criteria |
| 0:05 | 20  | PM hierarchy: daily / weekly / monthly / annual                      | Use handouts/<br>pm_checklists.md     |
| 0:25 | 15  | Documentation systems & fleet software overview                      | Paper vs digital; telematics mention  |
| 0:40 | 15  | Build-a-PM-plan tabletop (teams of 2)                                | 20-cart mixed fleet scenario          |
| 0:55 | 5   | Break                                                                |                                       |
| 1:00 | 20  | Career pathways: OEM training (Club Car, EZ-GO, Yamaha), shop growth |                                       |
| 1:20 | 25  | Program review game: rapid-fire 7-step, batteries, Curtis, safety    | Whiteboard scoring optional           |
| 1:45 | 10  | Case study presentation expectations & rubric                        |                                       |
| 1:55 | 5   | Written exam logistics for Session 2                                 |                                       |

### Key Teaching Points

- PM prevents the expensive roadside/fleet-down failures taught in Weeks 2–7.
- Lithium fleets still need inspection PM (connections, charger, software/BMS health) — different tasks, not “zero maintenance.”
- Certificates document competence — employers care about safety + process.

## Common Misconceptions

| Misconception                            | Correction                                       |
|------------------------------------------|--------------------------------------------------|
| "PM is just washing carts."              | Structured inspections, torques, tests, records. |
| "Annual PM is enough for rental fleets." | Daily/weekly checks catch unsafe carts early.    |
| "If it runs, skip documentation."        | Fleets live/die by records and repeatability.    |

## Safety Reminders

- Final practical still full PPE/LOTO rules — assessment conditions are real.
- No sharing answers on written exam.
- Road tests during practical only with clearance.

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## Session 2 — Case Studies, Finals & Certificates (2 hours)

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**Timing note:** If class >6, extend Session 2 or add a makeup block for practicals. Ideal practical is individual.

**Materials needed:** final\_assessment/practical\_skills\_rubric.md (1/student); final\_assessment/written\_final\_exam.md; secured answer key; certificates; faulted final carts; feedback forms; quiet written-exam space.

### Timed Agenda (template for 6-8 students)

| Time | Min | Activity                                             | Instructor Notes                                                   |
|------|-----|------------------------------------------------------|--------------------------------------------------------------------|
| 0:00 | 5   | Safety & exam integrity brief                        | Phones away for written                                            |
| 0:05 | 30  | Case study presentations (timed 4-5 min each)        | Score with documentation rubric                                    |
| 0:35 | 5   | Transition / water                                   |                                                                    |
| 0:40 | 45  | Final practical skills assessment (staggered starts) | Use practical rubric; critical items watched closely               |
| 1:25 | 30  | Written final exam (50 questions)                    | Quiet room; 30-40 min typical                                      |
| 1:55 | 5   | Collect exams; preview certificate/feedback timing   | Score offline if needed; award next day OR same day if graded fast |

**Alternate order (recommended if grading same day is required):** Written (30) → Presentations (25) → Practical (45) → Certificates/feedback (20). Adjust as needed; document chosen order in class notes.

## Certificate Award & Feedback

- Announce pass/fail privately if possible; celebrate publicly for passers.
- Collect program feedback (what to improve next cohort).
- Remind graduates: OEM manuals + Curtis code resources for lifelong learning.

## Assessment Focus

Case study quality, final practical performance, written score, attendance/participation. Students meeting all criteria receive **Golf Cart Diagnostic Technician** certificate of completion.

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## Week 10 Quiz (7 questions) — Optional mini-check before final; not a substitute for 50-question exam

1. Daily pre-use checks commonly include:
  - A) Full armature rebuild
  - B) Tires, obvious damage/leaks, functional brakes, unusual noises, basic battery condition cues
  - C) Controller reflash only
  - D) Transaxle overhaul
2. Annual/hourly major PM may include:
  - A) Ignoring batteries
  - B) Full inspection, battery bank load testing, controller diagnostics, transaxle service as scheduled
  - C) Removing all fuses permanently
  - D) Disabling HPD
3. True or False: Lithium carts require no inspections because they need no watering.
4. A case study presentation should include:
  - A) Symptom → process → data → diagnosis → repair → verification
  - B) Only the part invoice
  - C) Customer passwords
  - D) Unrelated stories only
5. Short answer: Name two OEM technician training brands referenced in this program.
6. Short answer: What attendance guideline supports certificate eligibility in the Program Guide?
7. Short answer: Why keep PM records in a fleet operation?

## Answer Key

1. **B**
2. **B**

3. **False** — still inspect connections, charger, BMS/health, mechanical systems
  4. **A**
  5. Club Car, EZ-GO (Textron), Yamaha (any two)
  6.  $\geq 85\%$  attendance or completed makeup labs (per Program Guide)
  7. Accountability, warranty/defense, trend spotting, scheduling, safety compliance
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## Instructor Post-Cohort Checklist

- ☐ Score practicals & written finals
- ☐ Issue certificates
- ☐ Update ITERATION\_LOG with after-action notes
- ☐ Reset all planted faults; restore carts to safe fleet condition
- ☐ Inventory tools/PPE damage